

Combinatorial chemistry yields a learnable transcriptomic design map in Z-Screen

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Abstract

Most phenotypic screens end as a list of active wells. Z-Screen was designed to leave behind something more reusable: a map from combinatorial chemistry provenance to measured RNA state. The platform profiles one-bead-one-compound libraries in 50,000-well microchips, with each microwell containing 5 to 10 cells and carrying both compound identity and a low-pass transcriptomic readout. Across 615,793 repaired RNA profiles spanning 12 combinatorial libraries and 4 cell lines (615,721 with valid scVI latent coordinates), the present pilot asks whether that RNA-first chemistry record is already structured enough to support prediction, candidate triage, and reuse.

The assay floor is high across the main systems. Median split-half cosine for control compounds is 0.993 in A549, 0.991 in H1650, 0.993 in HEK293, and 0.968 in THP1. On held-out tuples within a library, building-block models substantially outperform centroid baselines. The strongest case is ZEL024 in HEK293, where a building-block plus pair model reaches median cosine 0.674 against a 0.397 centroid baseline (33.2% reduction in mean squared error). ZEL031 in THP1 follows at cosine 0.833 vs 0.783 (21.3% error reduction). Chemistry-only descriptors carry predictive signal as well: in ZEL024 / HEK293, a combined building-block plus chemistry-embedding model reaches cosine 0.621 with a 29.3% error reduction, the highest gain among any chemistry-only configuration evaluated.

Full-structure descriptors add signal where the system is best sampled, but not uniformly. In ZEL024 / HEK293, a combined building-block plus chemistry-embedding model reaches cosine 0.621 with a 29.3% error reduction, exceeding either representation alone. In weaker or sparser settings, full-structure augmentation is neutral or harmful, which makes ZEL024 / HEK293 the current positive case rather than proof of a universal structure model. The same calibration applies to reference-state recovery. ZEL024 / HEK293 contains a library tuple near the internal reference ZF-104 at cosine 0.901 against a HEK293 cross-control background of median 0.200 and 99th percentile 0.931. ZEL031 / THP1 contains high-cosine neighborhoods near sorafenib, MZ1, and STC-15, but THP1 also has a higher cross-control background (median 0.589, 99th percentile 0.972), so these are candidate neighborhoods, not mechanism calls. Building-block effects recur across cell lines at chemistry-resolved positions, and imaging adds a smaller secondary benefit in additive benchmarks.

Together, the results support a constrained but useful claim: in the best-sampled Z-Screen libraries,

explicit chemistry provenance plus low-pass RNA learns reusable chemistry-resolved biological programs. Imaging is a secondary extension in the present package. The core asset today is the RNA-derived transcriptomic design map and the provenance that makes each measured state reusable.

Significance

The present analysis tests whether Z-Screen can convert a combinatorial screen into reusable transcriptomic memory. Because every compound carries building-block provenance and a measured RNA response, the dataset can ask which chemical pieces move cell state, which combinations are predictable, which new tuples occupy neighborhoods of known references, and which programs survive a change in cell line. The result is not a mechanism assignment engine. It is a calibrated design map that can prioritize chemistry for follow-up.

Introduction

Early-stage discovery needs two kinds of information at once. It needs scale, because most chemistry is inactive or uninteresting in a given context. It also needs enough biological resolution to make a hit useful after the primary screen. Imaging platforms provide scale and rich morphology, but morphology alone is often difficult to translate into a chemical design decision. Transcriptomic profiling is closer to pathway state, but it is usually too expensive to apply deeply across combinatorial chemistry. Public transcriptomic atlases such as LINCS L1000 and Tahoe-100M established the value of perturbational RNA at large scale [2,3], while public multimodal efforts such as scGeneScope show the value of aligning imaging and RNA across matched treatments [4].

The gap addressed here is chemistry provenance. A conventional compound atlas can tell a discovery team that one molecule resembles another in expression space. It usually cannot say which building block, linker, or combinatorial position contributed to that state, because the chemistry was not generated as a measured design grammar. Z-Screen starts from the opposite end. Each compound is a tuple from a one-bead-one-compound library, and each measured RNA response can be traced back to building-block coordinates and full compound structure. That makes the primary question practical: does the RNA readout contain enough reproducible, chemistry-resolved signal to predict held-out programs and guide the next synthesis decision?

The present manuscript treats imaging as a current extension rather than the main claim. Same-well morphology is valuable, and companion paper1 focuses on that readout. In this paper, the evidence chain is RNA-first: reproducibility establishes the assay floor, building-block provenance tests predictability, calibrated reference-state neighborhoods test discovery-style triage, cross-cell comparisons ask whether programs can be reused, and the final multimodal benchmark asks whether imaging adds signal on top of the RNA map.

The analyses below characterize what the canonical dataset, itself assembled from a small number of pilot experiments, currently supports. Six concrete questions are addressed:

1. Are control compounds reproducible enough to support modeling?

2. Does building-block composition predict held-out transcriptomic states?
3. Where does full chemical structure add predictive signal beyond building-block identity?
4. Do library tuples recover recognizable reference-state neighborhoods once cross-control similarity backgrounds are accounted for?
5. Do building-block effects recur across cell lines?
6. Does imaging contribute secondary value in a restrained multimodal benchmark?

Each question is answerable on the present pilot. The aim is to define what the current data already supports, where it is strongest, and which next analyses are required before broader claims about out-of-family chemical generalization or mechanism can be made.

Results

Control profiles establish a stable assay floor across cell lines

The chemistry-prediction question is only meaningful if the RNA readout is stable enough to support it. Split-half reproducibility for named controls is high across the main systems. Median control cosine is 0.993 in A549, 0.991 in H1650, 0.993 in HEK293, and 0.968 in THP1. THP1 sits below the others because its compound-control library coverage is sparser, not because the readout is weaker on monocytes.

These numbers establish the floor for the rest of the paper. They also leave headroom between technical reproducibility and model performance: a model reaching cosine 0.674 is not capped by an unstable assay. With reproducibility established, the next question is whether the chemistry provenance attached to each RNA profile is predictive.

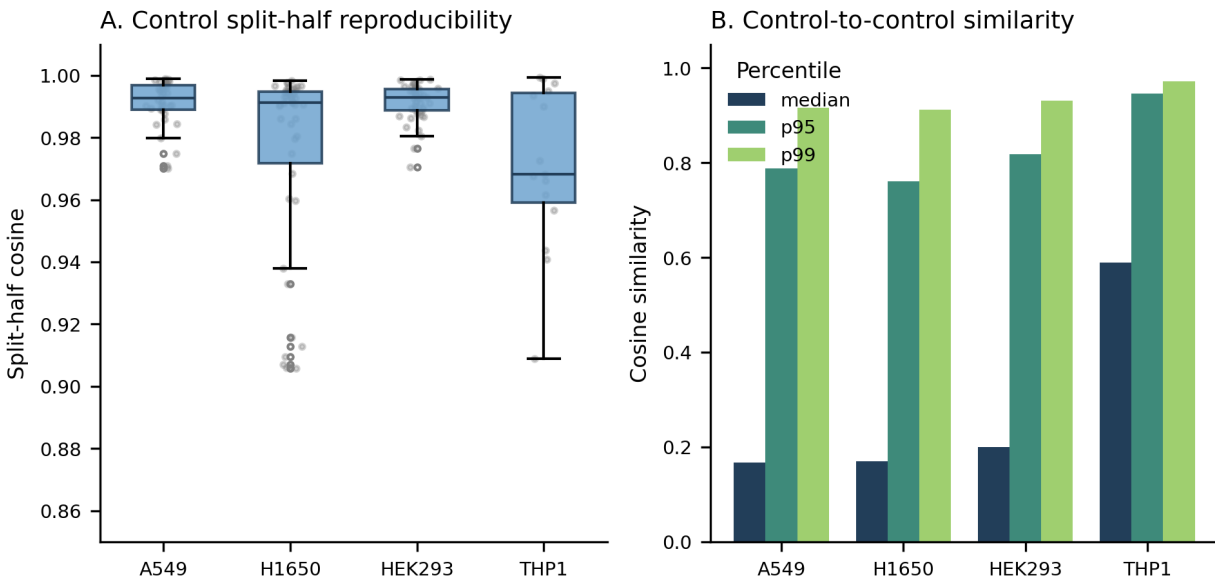


Figure 1: Named controls show high split-half reproducibility across all four cell lines, establishing a strong assay floor for downstream chemistry-to-phenotype modeling.

Building-block models predict held-out compound programs

The central modeling question is whether compositional chemistry carries enough signal to predict the transcriptomic state of a held-out library member. Across several library and cell-line systems, the answer is yes, with the strongest evidence coming from the best-sampled libraries.

ZEL024 in HEK293 is the strongest case. A building-block plus pair model reaches median cosine 0.674 against a 0.397 centroid baseline, a 33.2% reduction in mean squared error. ZEL031 in THP1 follows at cosine 0.833 vs 0.783 (21.3% error reduction). ZEL031 in A549 and ZEL024 in H1650 also showed positive gains, although less pronounced.

The model does more than memorize whole compounds within the tested split. It learns that particular chemical pieces and combinations of pieces carry recurring transcriptomic programs, which supports design-time reasoning about combinations not yet screened. The boundary is important: within the broader generalization-ladder framework laid out in companion paper³, this benchmark is interpolation within the shared building-block vocabulary of each library. Harder held-building-block and scaffold-family transfer regimes are evaluated separately there.

Once building-block provenance predicts RNA state, the next question is whether full chemical structure adds information beyond the library grammar.

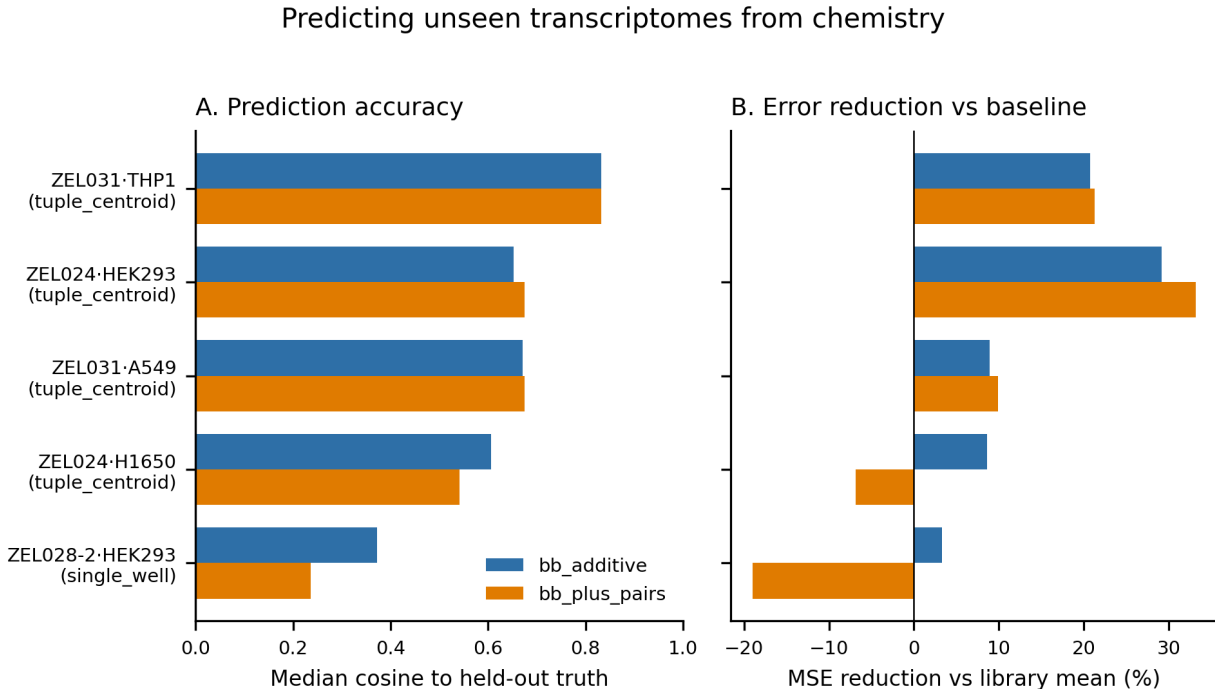


Figure 2: Building-block models improve prediction of held-out transcriptomic states across multiple library and cell-line systems, with the strongest gains in ZEL024 / HEK293 and ZEL031 / THP1.

Full chemical structure adds signal mainly in the strongest ZEL024 / HEK293 setting

Building-block identity is the strongest single signal in this dataset. Full structure adds further signal most clearly in ZEL024 / HEK293. In that system, a 256-dim chemistry embedding alone (a private random projection of ECFP4 fingerprints, see Methods) reached cosine 0.611 (27.8% error reduction), building-block identity alone reached 0.611 (27.5% error reduction), and combining the two reached 0.621 (29.3% error reduction).

That positive result should not be generalized to every current system. In ZEL024 / H1650, ZEL031 / A549, and ZEL031 / THP1, adding the full-structure embedding to building-block identity did not improve the mean error reduction over the building-block-only model. In sparse ZEL028-2 / HEK293, all chemistry-only variants performed below the mean-RNA baseline. The present conclusion is therefore narrow: full-structure descriptors can add design-time signal when coverage and assay context are favorable, with ZEL024 / HEK293 as the clearest example.

This hierarchy is useful rather than disappointing. Building-block provenance is the robust design coordinate today. Full structure is a candidate extension that becomes valuable when the measured library is dense enough for substructure information to be learned. Companion paper³ quantifies how much of the chemistry-only gain reflects interpolation within a library and how much survives stricter held-building-block evaluation. Having established predictability, the next question is whether the measured map can recover neighborhoods around known reference states.

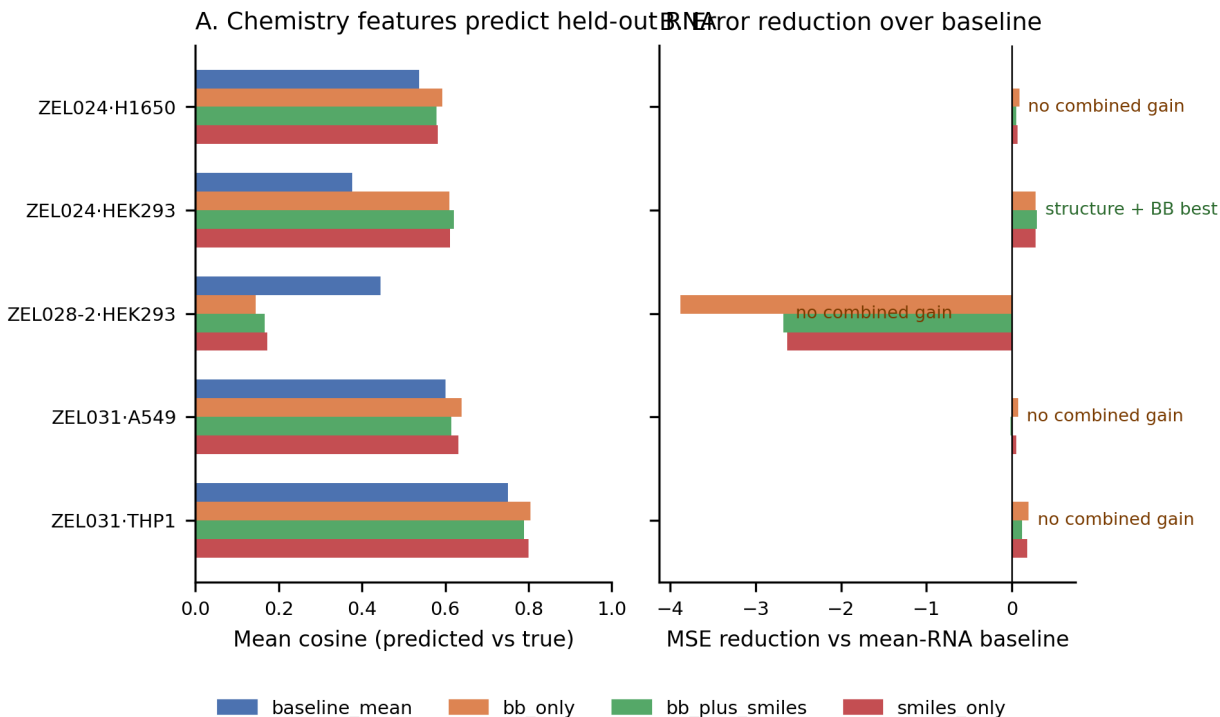


Figure 3: Chemistry-only models capture predictive signal unevenly across systems. Building-block provenance is the most reliable coordinate, and adding full-structure descriptors improves performance most clearly in ZEL024 / HEK293 while failing to improve or hurting in weaker and sparser settings.

Library tuples recover calibrated reference-state neighborhoods

The most direct discovery-style use of the map is not to prove mechanism, but to recover candidate neighborhoods around known reference compounds. A library tuple whose transcriptomic centroid lands near a reference centroid is worth follow-up only after calibration against the same-cell-line background of unrelated control-control similarities.

The cleanest neighborhood result comes from ZEL024 / HEK293, where a library tuple matched the internal reference ZF-104 at cosine 0.901. The HEK293 between-control similarity background sits at median 0.200 with a 99th percentile of 0.931. That places the 0.901 match far above the bulk of unrelated controls but still below the most extreme control-control tail. Several additional ZEL024 / HEK293 tuples reached cosine 0.85 to 0.89 against published references such as ZEL029-25 and ZEL029-28.

Candidate neighborhoods also appear in ZEL031 / THP1, with tuples reaching cosine 0.942 to sorafenib, 0.926 to MZ1, and 0.918 to STC-15. THP1 is useful for several pharmacology programs, but its cross-control similarity background is substantially higher (median 0.589, 99th percentile 0.972). The THP1 hits are biologically meaningful and populate the upper tail of plausible neighborhoods, but they should not be promoted as shared-mechanism evidence without orthogonal validation. The

THP1 background is reported explicitly so the calibration is visible to readers.

The defensible claim is calibrated candidate recovery. Z-Screen produces new chemistry that lands in recognizable RNA neighborhoods, and the strongest current case comes from a cell line where the cross-control background is low enough to make the neighborhood specific. These neighborhoods nominate follow-up experiments; they do not by themselves establish target engagement or mechanism. The next question is whether the chemistry programs behind these neighborhoods remain reusable when the cellular context changes.

Library tuples vs nearest reference compound

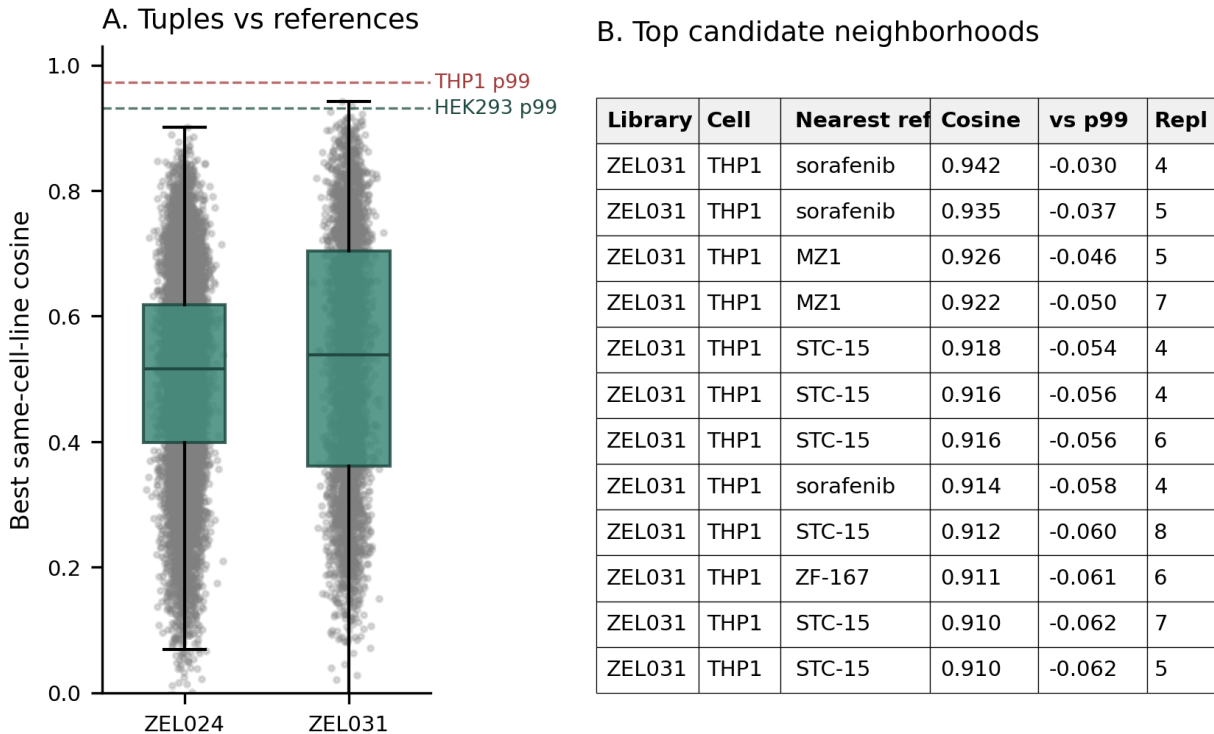


Figure 4: Figure 4. Multiple library tuples land near transcriptomic neighborhoods centered on known reference compounds. These are calibrated candidate neighborhoods, not mechanism assignments. The strongest current case is HEK293 (ZF-104 at cosine 0.901, against a HEK293 99th-percentile cross-control background of 0.931); THP1 hits to sorafenib, MZ1, and STC-15 occur in a system with a higher cross-control baseline.

Building-block programs recur across cell lines

A reusable chemistry model is more valuable when its building-block programs survive a change in cellular context. Per-building-block transcriptomic programs were compared in matched library systems across paired cell lines. Results are reported by building-block position because positions with very few unique values (e.g. ZEL024 bb0, bb1, bb2) collapse hundreds-to-thousands of distinct chemistries into a single average and therefore measure library-wide rather than chemistry-resolved

transfer. The chemistry-resolved positions are ZEL024 bb3 (83 building blocks, median 168 unique tuples averaged per signature) and ZEL031 bb0 / bb1 (a 2-position library, where each single-position signature still leaves the other position to vary across approximately 70 to 100 distinct partner chemistries).

Across these chemistry-resolved positions, building-block programs partially survive cell-context change. In ZEL024, the median cross-cell cosine at bb3 between HEK293 and H1650 was 0.602 ($n = 83$, IQR 0.51 to 0.69, strongest building blocks reaching 0.873). In ZEL031 between A549 and THP1, the median cross-cell cosine was 0.355 at bb0 ($n = 105$, IQR 0.23 to 0.48, strongest 0.747) and 0.526 at bb1 ($n = 84$, IQR 0.32 to 0.67, strongest 0.808).

These medians sit meaningfully below the within-cell reproducibility ceiling, so the result is not a claim of perfect transfer. The defensible interpretation is that many building blocks carry partially conserved biological directionality across cell types, which is the prerequisite for reusing chemistry across programs. The ZEL024 bb0 / bb1 / bb2 cross-cell results are not chemistry-resolved at this library architecture (each averaged signature reflects 900 to 7,000 distinct tuples) and are therefore reported as diagnostics in `paper2a/tables/bb_effect_consistency.csv` rather than as headline claims. Companion paper4 extends the cross-cell question to the molecule and grouped-program level, where the transfer story strengthens substantially in chemistry-resolved settings.

Cross-cell recurrence makes the RNA map more reusable, but it does not make imaging an equal pillar in this paper. The final Results analysis therefore asks a narrower question: whether the current image features add measurable signal on top of RNA in a simple additive benchmark.

Shared BB program consistency across cell lines

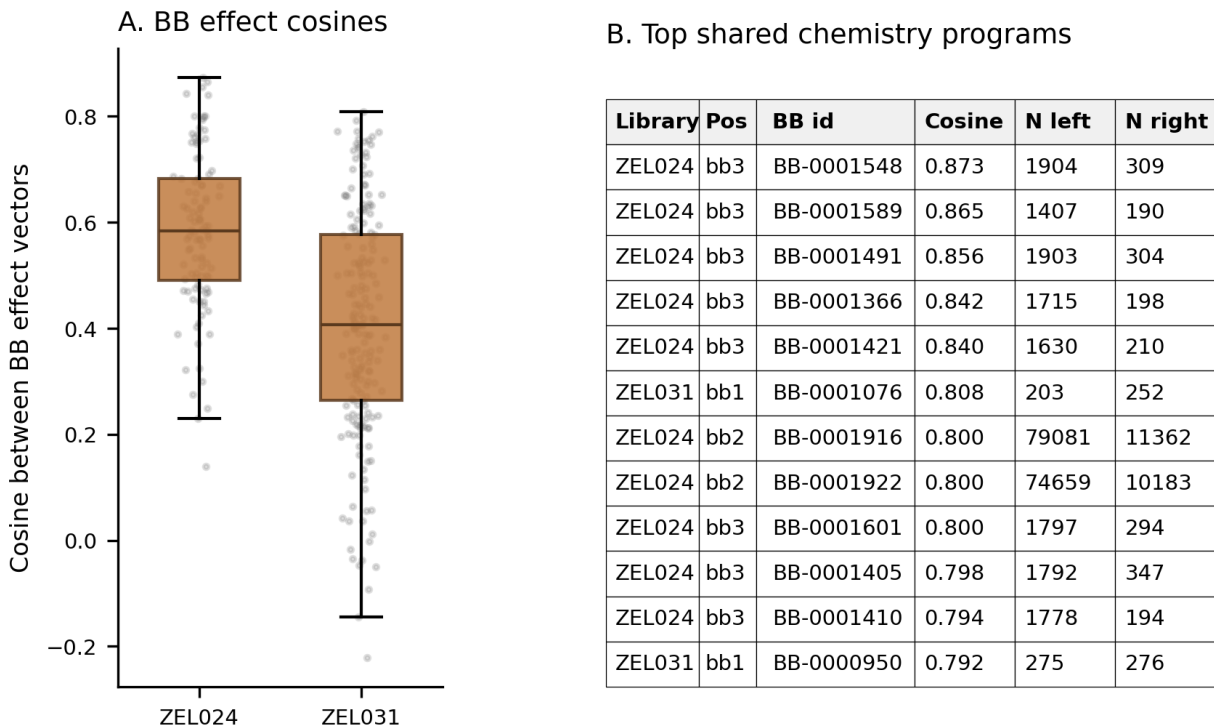


Figure 5: Building-block effects partially recur across cell lines at chemistry-resolved positions, with positive median transfer and a subset of building blocks showing strong cross-context conservation.

Imaging adds a modest secondary signal in a simple multimodal benchmark

The evidence in this paper is RNA-first, and the image branch is best read as a secondary extension in this analysis. In the strongest additive multimodal benchmark, based on ZEL024 controls, image-only classification reached 0.213 balanced accuracy, RNA-only reached 0.524, and image plus RNA reached 0.540. Adding chemistry to image plus RNA reached 0.611, consistent with the broader conclusion that provenance is the dominant non-RNA signal in this paper. The smaller ZEL031 recurrent-tuple benchmark followed the same ordering for image and RNA at lower absolute values: 0.127 for image, 0.146 for RNA, and 0.180 for image plus RNA. Chemistry-only and chemistry-containing models reach perfect accuracy in that small ZEL031 task, but that benchmark contains only 7 top-tuple classes with about 30 held-out examples and should be treated as a diagnostic rather than a broad multimodal claim.

The multimodal claim at the present pilot is measured. Imaging is not the lead modality today, but it adds directionally useful signal in an additive classification task. Companion paper1 shows a larger image contribution in the same-well ActiveSeq matched-readout setting after task-aware calibration of the per-well image latent, suggesting that the modest gain reported here reflects current image-pipeline maturity rather than a ceiling on same-well multimodality.

ActiveSeq-style additive multimodal probe

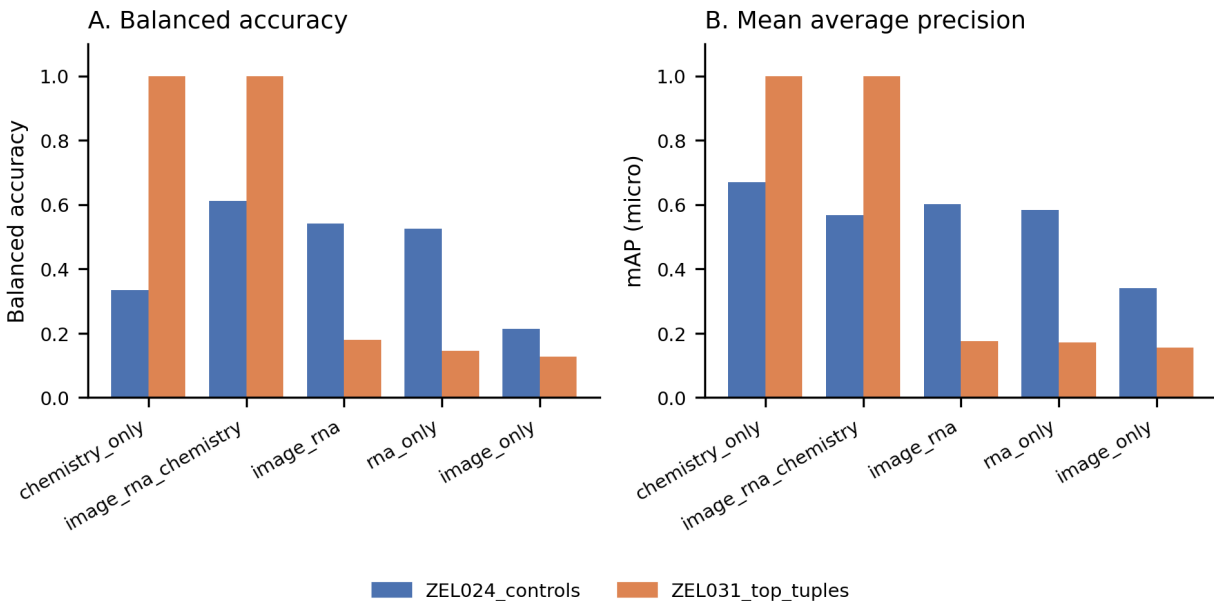


Figure 6: Figure 6. In a simple additive benchmark, image plus RNA modestly outperforms either single modality alone in ZEL024 controls, while RNA and chemistry provenance remain the dominant signals in the current dataset.

L1000 concordance provides an external validation check

The preceding analyses are internal to Z-Screen. A separate validation question is whether Z-Screen rank signatures resemble an independent transcriptomic platform where the same compounds and cell lines overlap. Of 47 named Z-Screen control compounds, 14 (29.8%) appear in the LINCS L1000 CMap LINCS 2020 build. Three of four Z-Screen cell lines (HEK293, A549, THP1) are present in LINCS L1000 with sample coverage; H1650 is absent.

For the concordance test, Z-Screen rank signatures were restricted to the 12,328-gene LINCS-visible universe so the two platforms were scored in the same gene space. Spearman rank correlation was then computed between Z-Screen up/down rank signatures and LINCS L1000 level-5 consensus signatures (NCBI GEO GSE70138 and GSE92742) for each adequately covered compound and cell-line pair. Across 11 such pairs (10 in A549, 1 in HEK293), all 11 produced positive cross-platform Spearman correlation. Median rho was 0.490, mean 0.393, and 8 of 11 pairs cleared $\rho = 0.2$. Five compounds reached empirical $p \leq 0.048$ against a same-cell-line unmatched-compound permutation null: ZM-336372 ($\rho = 0.625$, $p = 0.022$), rucaparib ($\rho = 0.530$, $p = 0.023$), sorafenib in A549 ($\rho = 0.505$, $p = 0.023$), veliparib ($\rho = 0.490$, $p = 0.046$), and crizotinib ($\rho = 0.679$, $p = 0.048$). Two compounds (palbociclib at $\rho = 0.182$, GSK126 at $\rho = 0.046$) fall in the bulk of the unmatched distribution; the GSK126 result is consistent with the documented slow transcriptional onset of EZH2 inhibition under LINCS’s 6 h treatment window.

This external comparison is not a benchmark victory claim. It is a sanity check that Z-Screen’s low-pass RNA signatures recover compound-specific transcriptional structure reported by an independent platform, using a gene-restricted comparison with no Z-Screen-specific calibration.

Discussion

Z-Screen behaves as a learnable chemistry-to-transcriptome system on data from a small number of pilot experiments. The evidence chain is intentionally sequential. Control reproducibility shows that low-pass RNA is stable enough for modeling. Building-block models show that explicit chemistry provenance predicts held-out transcriptomic programs. Reference-state analysis shows that library tuples can recover calibrated neighborhoods around known compounds. Cross-cell comparisons show that some building-block programs remain partially reusable across cellular contexts. Imaging then adds a modest secondary signal on top of the RNA-first map.

There is also a clear hierarchy in the current evidence. RNA carries the strongest biological signal today, and building-block provenance is the most reliable chemistry coordinate. Full-structure descriptors add signal in ZEL024 / HEK293, the best-sampled positive case, but they do not yet define a universal full-structure model across all libraries and cell lines. Imaging is directionally positive here and considerably stronger when calibrated for same-well matched readout in companion paper1. In this paper, the imaging branch is best framed as an additive layer on a working chemistry-to-RNA model.

The external position is therefore specific. Tahoe-100M is much larger on raw RNA scale, LINCS L1000 remains the deepest public small-molecule transcriptional catalog, scGeneScope is the closest public treatment-matched multimodal benchmark, and Cell Painting remains the standard high-throughput morphology assay [2-5]. Z-Screen is different because the chemistry is generated and measured as a compositional grammar. Every well carries building-block provenance, full compound identity, and an RNA response. The contribution is not raw atlas size; it is a design-relevant transcriptomic map in which a hit can be decomposed back into chemical choices.

The most informative next experiments follow directly. The first is greater depth in sparse libraries, especially systems that underperform when coverage per building block is low. The second is a prospective cross-library benchmark that asks whether a model trained on one scaffold family can predict transcriptomic programs in another; companion paper3 sets this up explicitly. The third is continued improvement in image consistency so that morphology can contribute more strongly to the readout. For a discovery team, the economic implication is that a screen run on this platform can leave behind a reusable model of chemistry-resolved RNA state rather than only a list of wells.

Methods

Dataset and preprocessing

The analysis uses the repaired canonical Z-Screen workspace, which contains RNA profiles, scVI latent coordinates, chemistry annotations, and selected image-derived features across 12 combinatorial libraries and 4 cell lines. Named controls were retained for reproducibility and reference-state neighbor-

hood analyses. Transcriptomic models were evaluated on library and cell-line subsets with sufficient compound coverage. The 72 ZIC004 / A549 rows without valid scVI latent coordinates were retained in the RNA count table but excluded from latent-space modeling; this is the source of the small mismatch between `canonical_obs.parquet` (615,793 rows) and `canonical_scvi_latents.parquet` (615,721 rows with valid coordinates) noted in the package README.

scVI latent space rather than gene-level expression

Most analyses operate on a 32-dimensional scVI latent space rather than full gene-level expression vectors, for two reasons. First, low-pass per-well mRNA-seq (each well is a pseudobulk of 5 to 10 cells) is dominated by sampling noise in any single gene; scVI’s variational latent absorbs that noise into shared structure across observations. Second, the latent is comparable across cell types to a degree that raw expression is not, which is the property the cross-cell transfer experiments in companion paper4 require. Where gene-level signal is the appropriate unit (reference-state neighborhood recovery, gene-program inspection in companion paper1) centered pseudobulk log-CPM is used directly.

Control reproducibility

Control split-half reproducibility was computed from repeated random partitions of wells belonging to the same reference compound. Similarity was summarized with cosine similarity between pseudo-replicate centroids in scVI latent space. Repeated partitioning rather than a single split was used to keep the variance of the metric low.

Building-block predictive modeling

Held-out tuple analyses fit additive and additive-plus-pair building-block models to transcriptomic centroids and evaluated them on compounds excluded from training by tuple identity. The choice of additive plus pair (rather than a high-order interaction model) reflects the present library coverage: pair-level interactions are well sampled in ZEL024, while higher-order terms run into combinatorial explosion at present scale. Performance was summarized with median cosine similarity to the measured transcriptomic target and mean squared error reduction relative to a centroid baseline. Companion paper3 extends this analysis to the harder building-block holdout and cross-library transfer regimes.

Chemistry-to-RNA prediction

Chemistry-only benchmarks compared models using a 256-dim per-compound chemistry embedding, building-block one-hot identities, or their concatenation. The chemistry embedding is a private fixed-seed Gaussian random projection from R^{2048} to R^{256} applied to per-compound 2048-bit ECFP4 (Morgan radius 2) fingerprints, then L2-normalized; this representation preserves Tanimoto-like geometry under cosine similarity (Johnson-Lindenstrauss) while being mathematically irreversible to substructure information without the projection matrix, which is held privately in the NDA bundle. A fixed deterministic projection was used in preference to a learned chemistry encoder

because the projection is reproducible across runs, requires no separate training step, and decouples the published chemistry features from the underlying SMILES strings. Outputs were scored against measured transcriptomic centroids using cosine similarity and mean squared error reduction relative to a centroid baseline. Full-structure claims were interpreted system by system rather than pooled across libraries.

Reference-state neighborhood recovery, cross-control normalization, and cross-cell building-block analysis

Reference-state neighborhood recovery identified library tuples whose transcriptomic centroids were most similar to named reference compounds within the same library and cell line. Hits were calibrated against the per-cell-line cross-control similarity background (median, 95th percentile, 99th percentile cosines among unrelated reference compounds in the same cell line), so that “high” similarity is interpreted relative to the cell-line baseline rather than against zero. The analysis nominates candidate neighborhoods for follow-up and was not used as proof of shared mechanism. Cross-cell building-block consistency compared the direction of per-building-block transcriptomic programs across paired cell lines within the same library.

Additive multimodal benchmark

The multimodal probe followed a conservative additive framing: image features, RNA features, and their concatenation were evaluated on held-out class prediction tasks constructed from the available image-linked data. Performance was summarized with balanced accuracy and mean average precision. The additive frame was chosen because it produces a result traceable to a specific feature axis; cross-attention or contrastive multimodal models can in principle do better, but the gain is harder to attribute to one modality.

External benchmark context

Tahoe-100M, scGeneScope, L1000, Cell Painting, and Z-Screen statistics were assembled into a comparison table covering RNA scale, image scale, cell-line coverage, perturbation diversity, pairing structure, and a representative result per dataset. The comparison was used only for Discussion-level positioning, not to claim parity on raw scale or multimodal balanced accuracy.

Limitations

The current analysis is based on a small number of pilot experiments rather than a fully saturated atlas. RNA carries the strongest signal today; the image branch is directionally positive and depends on derived feature artifacts rather than raw microscopy in this shareable package. Full-structure descriptors add clear signal mainly in ZEL024 / HEK293 and should not yet be treated as a universal structure-to-RNA model. Reference-state neighborhoods are calibrated against same-cell-line cross-control backgrounds, but even the strongest discovery-style hits require orthogonal experimental validation before they should be treated as confirmed mechanisms of action. Cross-library and

prospective scaffold-family transfer are set up by the present results but are not claimed as solved here.

Data availability

Data tables, derived image features, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper2a/`, `data/ZScreen_Canonical_Dataset/`, and `data/paper2_artifacts/`. Raw microscopy images are not included in the shareable bundle; the paper2a image-linked analyses use the derived parquet feature files documented in the package README. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper2a/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

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